

PAPER #37 F_UBii Buoyancy Force: Proof Variants 26 (Thermodynamic Series)

Title: UQFF Buoyancy Proof Variants 26: Terminal Velocity, Ionization, Energy Coupling, Orbital Decay, and Kilonova Buoyancy

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Grok Thread: 98b2e77dfbc34d27b09f19fa7c460624

Validator: BuoyancyProofVariants.py All 17 variants operational ?

Variants: termv, upar, coup, orbdec, kn

Index Slot: 1.5 Buoyancy Proofs, Paper #37

Abstract

This paper presents five F_UBii buoyancy proof variants spanning thermodynamic processes from jet terminal velocities to kilonova ejecta. Variant 2 (termv) applies to astrophysical jets and winds reaching terminal velocity balance. Variant 3 (upar) addresses photoionized regions governed by the ionization parameter U . Variant 4 (coup) quantifies energy coupling efficiency in accretion disk and reconnection contexts. Variant 5 (orbdec) derives the buoyancy analog of gravitational wave-driven orbital decay in compact binaries. Variant 6 (kn) applies to the kilonova AT2017gfo, predicting $F_{\text{UBii_kn}} = 1.3051054 \text{ N}$ for ejecta with $L_{\text{peak}} = 5104 \text{ W}$, $t_{\text{peak}} = 1 \text{ day}$, and $M_{\text{ej}} = 0.05 \text{ M}_{\odot}$. Together, these five variants form the thermodynamic series of the F_UBii taxonomy.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.01074 \text{ day}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Variant 2: Terminal Velocity Jet/Wind Buoyancy (termv)

1.1 Physical Context

Astrophysical jets (AGN, GRB, protostellar) and stellar winds accelerate material to a terminal velocity v_{term} where radiation pressure, magnetic driving, and gravitational drag balance. At terminal velocity, $da/dt = 0$ and the net buoyancy force represents the frozen-in momentum of the outflow.

Key systems: M87 jet ($v_{\text{term}} \sim 0.98c$), Sgr A* winds ($v_{\text{term}} \sim 0.1c$), OB stellar winds ($v_{\text{term}} \sim 10003000 \text{ km/s}$)

1.2 F_UBii_termv Equation

$$F_{\text{UBii,termv}} = F_{\text{rel}} \cdot \frac{\tau \cdot L}{c \cdot E_{\text{LEP}}} \cdot Q_{\text{wave}} \cdot v_{\text{term}}$$

where: - τ = optical depth / momentum transfer timescale (s) - L = jet/wind luminosity (W) - v_{term} = terminal velocity (m/s)

1.3 Physical Derivation

The momentum flux of a radiation-driven wind:

$$\dot{p} = \frac{\tau L}{c}$$

The UQFF buoyancy enters through the E_{LEP} normalization the lepton energy scale sets the quantum granularity of momentum transfer:

$$F_{\text{UBii,termv}} = \dot{p}_{\text{UQFF}} \cdot v_{\text{term}} = F_{\text{rel}} \cdot \frac{\tau L}{c \cdot E_{\text{LEP}}} \cdot Q_{\text{wave}} \cdot v_{\text{term}}$$

1.4 Example: M87 Relativistic Jet

For M87 jet: $\tau = 10^3$, $L = 10^{44}$ W, $v_{\text{term}} = 0.98c = 2.94108 \times 10^8$ m/s, $Q_{\text{wave}} = 1.0$:

$$F_{\text{UBii,termv}}^{M87} = 10^{-10} \times \frac{10^{-3} \times 10^{44}}{3 \times 10^8 \times 1.22 \times 10^{-19}} \times 2.94 \times 10^8 = 10^{-10} \times 2.73 \times 10^{48} \times 2.94 \times 10^8 = 8.0 \times 10^{46}$$

This represents the UQFF momentum-flux buoyancy of the M87 jet the force that keeps the relativistic plasma buoyantly confined against the ICM pressure of the Virgo Cluster.

2. Variant 3: Ionization Parameter Buoyancy (upar)

2.1 Physical Context

The ionization parameter $U = n_{\text{photons}}/n_{\text{H}}$ quantifies the ratio of ionizing photon density to hydrogen density. In HII regions, AGN narrow-line regions, and quasar broad-line regions: $U \sim 10^{-4}$ to 10^3 . This dimensionless ratio controls all ionic fractions and hence the buoyancy of photoionized gas.

2.2 F_UBii_upar Equation

$$F_{\text{UBii,upar}} = -F_{\text{rel}} \cdot \frac{U \cdot n_{\text{H}} \cdot r^2}{E_{\text{LEP}}} \cdot Q_{\text{wave}} \cdot \sqrt{U}$$

where: - U = ionization parameter (dimensionless) - n_{H} = hydrogen number density (m⁻³) - r = distance from ionizing source (m)

The negative sign reflects compression: photoionized gas is over-pressured by the radiation field and compresses surrounding neutral gas.

2.3 $U^{(3/2)}$ Scaling

The $F_{UBii_upar} \sim U^{(3/2)}$ scaling arises because: - Factor U : radiation pressure scaling - Factor vU : thermal pressure response ($T_e \propto U^{(1/2)}$ in ionized gas)

This gives $F_{UBii_upar} \propto U^{(3/2)} n_H r$ exactly the ram pressure of the HII region expansion front against surrounding neutral gas.

2.4 Example: Orion Nebula (M42)

For M42: $U \sim 10^7$, $n_H \sim 10^4 \text{ m}^{-3}$, $r \sim 3107 \text{ m}$ (1 pc), $Q_{wave} = 1.0$:

$$F_{UBii,upar}^{M42} = -10^{-10} \times \frac{10^{-2} \times 10^9 \times (3 \times 10^{17})^2}{1.22 \times 10^{-19}} \times \sqrt{10^{-2}} = -10^{-10} \times 7.38 \times 10^{45} \times 0.1 = -7.4 \times 10^{35}$$

This inward compression force (7.4105 N) represents the photoionization pressure confining the Orion Nebula's ionization front.

3. Variant 4: Energy Coupling Efficiency Buoyancy (coup)

3.1 Physical Context

Energy coupling efficiency $e_{coup} = E_{deposited}/E_{input}$ quantifies how efficiently energy input (from AGN, SNe, cosmic rays) couples to surrounding gas. In AGN feedback: $e_{coup} \sim 0.05$. In SNe: $e_{coup} \sim 0.1$. In magnetic reconnection: $e_{coup} \sim 0.01$.

3.2 F_{UBii_coup} Equation

$$F_{UBii,coup} = F_{rel} \cdot \frac{\varepsilon_{coup} \cdot \dot{E}}{E_{LEP}} \cdot Q_{wave} \cdot \sqrt{\varepsilon_{coup}}$$

where: - e_{coup} = energy coupling efficiency (01) - E = energy transfer rate (W)

3.3 $e^{(3/2)}$ Coupling Law

The $F_{UBii_coup} \propto e^{(3/2)} E$ scaling reflects the UQFF energy cascade: at high coupling efficiency, the buoyancy force scales super-linearly with coupling a physical manifestation of the non-linear positive feedback in AGN mechanical feedback.

3.4 Example: AGN Kinetic Feedback

For a radio-mode AGN: $e_{\text{coup}} = 0.05$, $E = 1044 \text{ W}$, $Q_{\text{wave}} = 1.0$:

$$F_{\text{UBii,coup}}^{\text{AGN}} = 10^{-10} \times \frac{0.05 \times 10^{44}}{1.22 \times 10^{-19}} \times \sqrt{0.05} = 10^{-10} \times 4.10 \times 10^{53} \times 0.2236 = 9.2 \times 10^{43} \text{ N}$$

4. Variant 5: Orbital Decay Binary Buoyancy (orbdec)

4.1 Physical Context

Compact binary systems (NS-NS, BH-BH, NS-BH) lose energy to gravitational wave radiation. The Peters formula gives:

$$\frac{da}{dt} = -\frac{64}{5} \frac{G^3 M_1 M_2 (M_1 + M_2)}{c^5 a^3}$$

The UQFF buoyancy analog replaces the pure GW energy loss with a field-theoretic force:

4.2 F_UBii_orbdec Equation

$$F_{\text{UBii,orbdec}} = -F_{\text{rel}} \cdot \frac{64}{5} \cdot \frac{G^3 M_1 M_2 (M_1 + M_2)}{c^5 \cdot a^4 \cdot E_{\text{LEP}}} \cdot Q_{\text{wave}} \cdot \frac{da}{dt}$$

where: - M_1, M_2 = component masses (kg) - a = semi-major axis (m) - da/dt = orbital decay rate (m/s)

The negative sign indicates inspiral the buoyancy force drives the binary inward.

4.3 Connection to Peters Formula

The Peters orbital decay rate (da/dt) enters linearly $F_{\text{UBii,orbdec}}$ is the UQFF field force per unit of GW power radiated:

$$F_{\text{UBii,orbdec}} = \frac{F_{\text{rel}}}{E_{\text{LEP}}} \cdot P_{\text{GW, Peters}} \cdot Q_{\text{wave}} \cdot |da/dt|$$

where $P_{\text{GW,Peters}}$ is the Peters formula for GW power. This establishes a direct UQFFGW correspondence.

4.4 Example: GW170817 Pre-Merger

For GW170817 (NS-NS): $M_1 = M_2 = 1.4 M_{\odot} = 2.78510 \text{ kg}$, $a = 2108 \text{ m}$ (final orbit), $da/dt = -10 \text{ m/s}$:

$$F_{\text{UBii,orbdec}} = -10^{-10} \times 12.8 \times \frac{(6.674 \times 10^{-11})^3 \times (2.785 \times 10^{30})^3}{(3 \times 10^8)^5 \times (2 \times 10^8)^4 \times 1.22 \times 10^{-19}} \times 10 = -10^{-10} \times 4.1 \times 10^{56} \times 1$$

5. Variant 6: Kilonova Peak Luminosity Buoyancy (kn)

5.1 Physical Context

Kilonovae are radioactively powered transients following neutron star mergers. AT2017gfo (counterpart to GW170817) achieved $L_{\text{peak}} \sim 5104$ W at $t_{\text{peak}} \sim 1$ day, with ejecta mass $M_{\text{ej}} \sim 0.05 M_{\odot}$. The r-process nucleosynthesis in the ejecta generates heavy elements (gold, platinum, uranium) through neutron capture.

5.2 $F_{\text{UBii_kn}}$ Equation

$$F_{\text{UBii,kn}} = F_{\text{rel}} \cdot \frac{L_{\text{peak}} \cdot t_{\text{peak}}}{E_{\text{LEP}}} \cdot Q_{\text{wave}} \cdot \left(\frac{M_{\text{ej}}}{M_{\odot}} \right)^{1/3}$$

where: - L_{peak} = peak bolometric luminosity (W) - t_{peak} = time to peak (s) - M_{ej} = ejecta mass (kg)

The $M_{\text{ej}}^{1/3}$ factor reflects the geometric (volumetric) scaling of the ejecta opacity.

5.3 AT2017gfo Calculation

For AT2017gfo: - $L_{\text{peak}} = 5104$ W - $t_{\text{peak}} = 86400$ s (1 day) - $M_{\text{ej}} = 0.05 M_{\odot} = 0.05 \cdot 1.98910 \times 10^{30} = 9.945108 \times 10^{28}$ kg - $Q_{\text{wave}} = 1.0$

$$F_{\text{UBii,kn}}^{\text{AT2017gfo}} = 10^{-10} \times \frac{5 \times 10^{40} \times 86400}{1.22 \times 10^{-19}} \times 1.0 \times (0.05)^{1/3}$$

- Numerator: $5104 \cdot 86400 = 4.321045 \times 10^8$
- Ratio: $4.321045 / 1.2210^{19} = 3.541064 \times 10^{-11}$
- F_{rel} : 3.541054×10^{-11}
- $(0.05)^{1/3} = 0.368: = 1.3051054 \times 10^{-11}$ N

$$F_{\text{UBii,kn}}^{\text{AT2017gfo}} = 1.305 \times 10^{54} \text{ N}$$

Validator confirms: BuoyancyProofVariants.py ? $F_{\text{UBii_kn}} = 1.3051054 \text{ N}$?

5.4 Physical Interpretation

The kilonova buoyancy force $F_{\text{UBii_kn}} = 1.3051054 \text{ N}$ represents the UQFF unified field response to the instantaneous energy release of the r-process. Comparison to the gravitational confinement force of the merger remnant:

$$F_{\text{grav}}^{\text{merger}} = \frac{G(M_1 + M_2)^2}{R_{\text{merger}}^2} \approx \frac{6.674 \times 10^{-11} \times (5.57 \times 10^{30})^2}{(10^4)^2} = 2.1 \times 10^{36} \text{ N}$$

The ratio $F_{UBii_kn} / F_{grav} = 1.3051054 / 2.1106 = 6.2107$ the UQFF kilonova buoyancy vastly exceeds gravitational confinement, explaining the explosive ejecta dynamics observed in AT2017gfo.

6. Summary: Thermodynamic Series F_{UBii} Values

Variant	Physical Context	Key Parameters	F_{UBii}
termv	M87 jet terminal velocity	$t=10?$, $L=1044$ W, $v_{term}=0.98c$	~ 81047 N
upar	Orion Nebula ionization	$\bar{U}=10?$, $n_H=10?$ m?, $r=1$ pc	~ 7105 N
coup	AGN kinetic feedback	$e=0.05$, $E=1044$ W	~ 9104 N
orbdec	GW170817 final orbit	$1.4+1.4$ M?, $a=200$ km	~ 41047 N
kn	AT2017gfo kilonova	$L=5104$ W, $M_{ej}=0.05$ M?	1.3051054 N ? validator ?

Conclusions

The thermodynamic series of F_{UBii} variants (26) demonstrates the versatility of the UQFF buoyancy framework across five distinct physical regimes:

1. **termv**: Connects UQFF to radiation-driven momentum flux in relativistic jets
2. **upar**: Maps photoionization pressure to UQFF energy scale via $U^{(3/2)}$ scaling
3. **coup**: Establishes non-linear AGN feedback response through $e^{(3/2)}$ coupling law
4. **orbdec**: Provides UQFF field-theoretic interpretation of Peters formula GW in-spiral
5. **kn**: Predicts AT2017gfo buoyancy $F = 1.3051054$ N validated by BuoyancyProof-Variants.py

All five variants share the common $F_{UBii} = F_U - F_{Bi} - F_i$ architecture with $F_{rel} = 10?$ N normalization and $E_{LEP} = 1.2210??$ J quantum granularity (Paper #36).

Validator: *BuoyancyProofVariants.py* ? All 17 F_{UBii} variants operational ? | ? = 0.0005/day | $[SSq] = 0.57$

See
also:
PA-
PER_036
|
Part
of
the
Star-
Magic
UQFF
Whitepa-
per
Se-
ries.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	5.0e-4 day ⁻¹	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10 ⁻²²	Inertia tensor scale
$E_{\text{react}}(0)$	10 ⁴⁶ J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 igl[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()

Term	Description	Implementation
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\Sigma \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconducting	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.